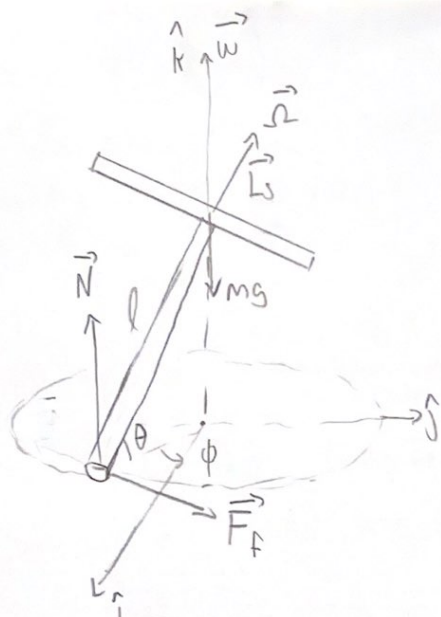


1)



Consider again the setup we've encountered in the last homework. This time we release the top with some initial spin angular velocity Ω_0 . Until the tip (of the top) starts rolling without slipping, it will slip for a while.

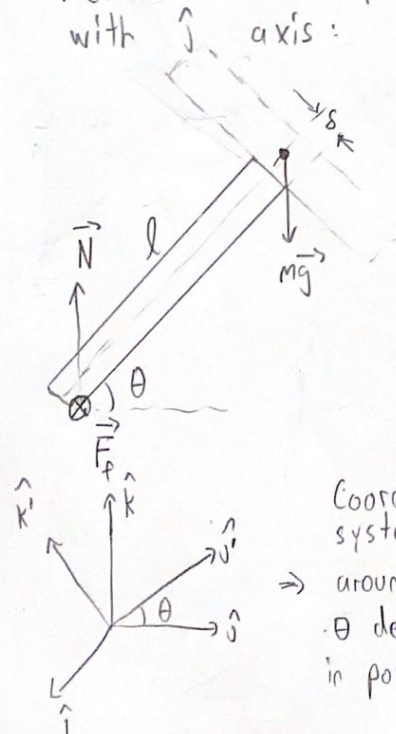
The friction acting on the top will create a torque and therefore our previous analysis is overruled.

We know that (from the previous homework) that for top to roll without slip at a certain angle θ , we need:

$$\Omega_c = \frac{l \cos \theta}{a} \sqrt{\frac{2g}{\delta}}$$

So for top to roll without slipping we need $\Omega_0 \geq \Omega_c$, otherwise it will eventually fall over or stabilize at a lower θ .

Let's write torque w.r.t. CM when the bottom segment of the top is aligned with $\hat{j}$ axis:



Coordinate system rotated around $\hat{i}$ by θ degrees in positive direction

$$\vec{r} = -l \hat{j}' - \delta \hat{k}' \quad : \text{position of contact point w.r.t to CM}$$

$$\vec{\tau} = \vec{r} \times \vec{F}_f + \vec{r} \times \vec{N}$$

$$= (-l \hat{j}' - \delta \hat{k}') \times F_f \hat{i} + (-l \hat{j}' - \delta \hat{k}') \times N \hat{k}$$

$$= \underbrace{F_f l \hat{k}'}_{\vec{\tau}_{\text{rise}}} - \underbrace{F_f \delta \hat{j}'}_{\vec{\tau}_{\text{spin}}} + \vec{\tau}_N$$

$$\vec{\tau} = \vec{\tau}_{\text{rise}} + \vec{\tau}_{\text{spin}} + \vec{\tau}_N$$

Now we have 3 main components of the torque:

$\vec{\tau}_N$ is basically responsible for the precession of the top around $\hat{k}$ as we have studied in the last homework.

But this time top is not rolling without slipping. It rather slips for a short time untill eventually Ω reaches $\Omega_c = \frac{L \cos \theta}{a} \sqrt{\frac{2g}{\delta}}$. There are more interesting motions this top does!

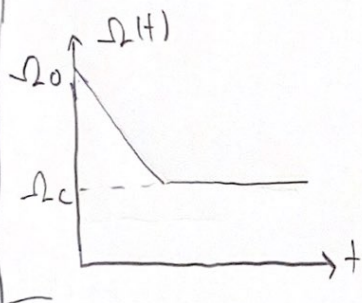
$\vec{\tau}_{spin}$ is in the direction of the symmetry axis. However it's opposing $\vec{L}_s$, spin angular momentum of the top. We know that

$$\tau_{spin} = \frac{dL_s}{dt} = I_3 \frac{d\Omega(t)}{dt} = -F_f \delta \Rightarrow \text{indep. of time}$$

$\propto t$

$$\Omega(t) = \Omega_0 - \frac{F_f \delta}{I_3} t = \Omega_0 - \frac{\mu M g \delta}{I_3} t \Rightarrow$$

We can assume $N \approx mg$ (although not exactly true).
Therefore $F_f = \mu N = \mu mg$



This relationship holds untill Ω reaches Ω_c . After that point, top will start rolling without slipping and it's spin will stay constant since friction is no more doing work on the top.

τ_{rise} is in $\hat{k}$ direction perpendicular to L_s . As in precession this torque will serve to change direction of L_s rather than it's magnitude.

$$\vec{\tau}_{rise} = \frac{d\vec{L}_s}{dt} = \vec{\omega}_\theta \times \vec{L}_s \quad \text{since} \quad \left(\frac{d\vec{L}}{dt} \right)_i = \left(\frac{d\vec{L}}{dt} \right)_{rot} + \Omega \times \vec{L}$$

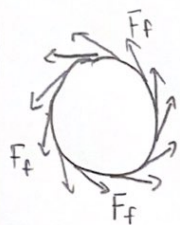
Here $\vec{\omega}_\theta$ is angular speed in $\hat{i}$ or θ direction.

This means that the top will actually rise up! ★

It's symmetry axis will move towards the vertical.



What happens when symmetry axis is along the vertical?
 Well, it depends on the shape of the peg that touches the floor. If it's cylindrical, friction force will not have a certain direction but rather will be circular.



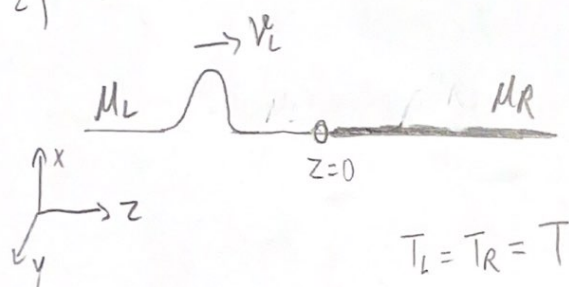
$\Rightarrow$ Top view of the peg when the whole surface is touching the floor $\theta = 90^\circ$

Therefore $\vec{F}_{rise}$ will be zero and top will not roll over to the other side and stay vertical.

The case where the peg is spherical is not so simple. The motion of the top will include nutation and the variation of $\vec{\omega}$ (precession angular velocity). Top can also roll over to the other side and is subjected to more complex motions.

For our purposes our analysis is satisfactory. We should realize that top will rise up and its spin will decrease until $\Omega = \Omega_c$. For a more detailed analysis one can check the article "How Does a Symetric Top Rise?" by V. Tarriverdi.

2)



String under homogeneous tension T with linear density obeying:

$$\mu(z) = \begin{cases} \mu_L, & z < 0 \\ \mu_R, & z > 0 \end{cases}$$

$$T_L = T_R = T$$

Wave equation $\mu(z) \frac{\partial^2 f}{\partial t^2} - T \frac{\partial^2 f}{\partial z^2} = 0$ clearly implies

$f_L(0, t) = f_R(0, t)$ and $f'_L(0, t) = f'_R(0, t)$ since string is continuous at $z=0$. We have:

$$\left. \begin{aligned} f_i(k_i'z - \omega_i t) &= f_i(z - v_L t) \Rightarrow \text{incident pulse} \\ f_R(-k_R'z - \omega_R t) &= f_R(z + v_L t) \Rightarrow \text{reflected pulse} \\ f_T(k_T'z - \omega_T t) &= f_T(z - v_R t) \Rightarrow \text{transmitted pulse} \end{aligned} \right\} \begin{aligned} v_L &= \frac{\omega_i}{k_i'} = \frac{\omega_R}{k_R} \\ v_R &= \frac{\omega_T}{k_T'} \end{aligned}$$

Continuity of $f(z, t)$ at $z=0$ requires:

$$f_i(-\omega_i t) + f_R(-\omega_R t) = f_T(-\omega_T t)$$

↳ For this to be true at all times, we must have $\omega_i = \omega_R = \omega_T = \omega$

One may think that ω is not meaningful in the context of pulses because they are not periodic and do not have frequencies associated with them,

HOWEVER, one may envision any form of wave (except those which have discontinuities) as superposition of infinitely many weighted and shifted sinusoidal waves.

$$f(z, t) = \int_0^\infty \gamma_\alpha \sin(k_\alpha z - \omega_\alpha t + \phi_\alpha) d\alpha$$

where γ_α is the weight and ϕ_α is the phase of the sinusoidal

with frequency ω . This is basically a primitive form of the Fourier transform.

One does not necessarily need to calculate weight and phases of the sinusoids. We only need to realize that:

$$\frac{\partial f}{\partial z} = k \int_0^{\infty} \gamma_{\alpha} \cos(kz - \omega t + \phi_{\alpha}) d\alpha$$

Also realize the relation we have found corresponds to:

$$f_i(-\omega^i t) + f_R(\omega^R t) = -f_T(-\omega^T t) \Rightarrow f_i + f_R = f_T$$

where f denotes the magnitude of the pulse. (Actually it's a measure of γ_{α} 's since form of the pulses won't change)

Also from the continuity of $f'(z, t)$ at $z=0$.

$$\left. \frac{\partial f_i(k^i z - \omega^i t)}{\partial z} \right|_{z=0} + \left. \frac{\partial f(-k^R z - \omega^R t)}{\partial z} \right|_{z=0} = \left. \frac{\partial f(k^T z - \omega^T t)}{\partial z} \right|_{z=0}$$

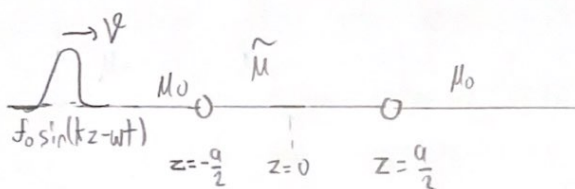
$$k^i \int_0^{\infty} \gamma_{\alpha}^i \cos(-\omega^i \alpha t + \phi_{\alpha}) d\alpha - k^R \int_0^{\infty} \gamma_{\alpha}^R \cos(-\omega^R \alpha t + \phi_{\alpha}) d\alpha = -k^T \int_0^{\infty} \gamma_{\alpha}^T \cos(\omega^T \alpha t + \phi_{\alpha}) d\alpha$$

That is $k^i f_i - k^R f_R = k^T f_T$

Recall that $k^i = k^R = \frac{\omega}{v_L}$, $k^T = \frac{\omega}{v_R}$

$$\begin{cases} f_i - f_R = \frac{v_L}{v_R} f_T \\ f_i + f_R = f_T \end{cases} \Rightarrow \begin{cases} 2f_i = \frac{v_L + v_R}{v_R} f_T \Rightarrow f_i = \frac{v_L + v_R}{2v_R} f_T \\ f_R = f_T - f_i \Rightarrow f_R = \frac{v_R - v_L}{v_L + v_R} f_i \end{cases}$$

3)



String under homogeneous tension T with linear mass density obeying:

$$\mu(z) = \begin{cases} \mu_0 & , z < -a/2 \\ \tilde{\mu} & , -a/2 < z < a/2 \\ \mu_0 & , z > a/2 \end{cases}$$

* One sends an infinite chain of monofrequency sinusoidal waves as $f_0 \sin(kz - \omega t)$ from $-\infty$.

* We can choose to observe the string much later than the moment one starts to send the sinusoidal waves so that the whole system has reached a steady state. This means all of the reflections and transmissions has occurred, and wave function is no more dependent on "t".

→ Instead of working with sinusoidal solutions, we can express waves as complex numbers. After doing all the calculations we should take the imaginary part of the complex exponential:

$$\text{Incident wave: } f_0 e^{i(kz - \omega t)} = f_i(z, t)$$

$$\text{Transmitted from the first interface: } \propto e^{i(\tilde{k}z - \omega t)} = f_{T-1}(z, t)$$

$$\text{Reflected from the first interface: } f_R e^{i(kz - \omega t)} = f_{R-1}(z, t)$$

$$\text{Reflected from the second interface: } \beta e^{i(-\tilde{k}z - \omega t)} = f_{R-2}(z, t)$$

$$\text{Transmitted from the second interface: } f_T e^{i(kz - \omega t)} = f_{T-2}(z, t)$$

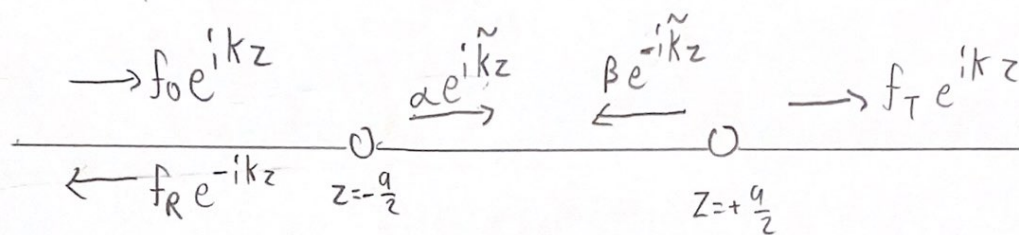
↳ Notice that wave number "k" for $f_{T-1}(z, t)$ and $f_{R-2}(z, t)$ are changed to $\tilde{k}$ because they are travelling in the segment of the rope where mass density is $\tilde{\mu}$.

We also reserved k for the segments of the rope having mass density μ_0 .

Now since we are observing the steady state (indep. of "t") we can drop the wt term in the exponential.

↳ OR one can take alternatively $t=0$ since it doesn't matter at which time we observe the waves.

We know from our analysis in class that wave function and its derivative must be continuous at interface since rope is essentially a construct of individual chains linked together.



From continuity of $f(x,t)$ and $f'(x,t)$ at $z = -\frac{a}{2}$

$$f_0 e^{ik\frac{a}{2}} + f_R e^{ik\frac{a}{2}} = \alpha e^{-i\tilde{k}\frac{a}{2}} + \beta e^{i\tilde{k}\frac{a}{2}} \quad (1) \Rightarrow f_L(x,t) = f_R(x,t)$$

$$f_0 k e^{-ik\frac{a}{2}} - f_R k e^{ik\frac{a}{2}} = \alpha \tilde{k} e^{-i\tilde{k}\frac{a}{2}} - \beta \tilde{k} e^{i\tilde{k}\frac{a}{2}} \quad (2) \Rightarrow f'_L(x,t) = f'_R(x,t)$$

From the continuity of $f(x,t)$ and $f'(x,t)$ at $z = \frac{a}{2}$

$$\alpha e^{+i\tilde{k}\frac{a}{2}} + \beta e^{-i\tilde{k}\frac{a}{2}} = f_T e^{ik\frac{a}{2}} \quad (3)$$

$$\alpha \tilde{k} e^{i\tilde{k}\frac{a}{2}} - \beta \tilde{k} e^{-i\tilde{k}\frac{a}{2}} = f_T k e^{ik\frac{a}{2}} \quad (4)$$

→ Multiplying (1) with k and summing with (2)

$$2 k f_0 e^{-ik\frac{a}{2}} = \alpha (k + \tilde{k}) e^{-i\tilde{k}\frac{a}{2}} + \beta (k - \tilde{k}) e^{i\tilde{k}\frac{a}{2}}$$

→ Multiplying (3) with $-k$ and summing with (4)

$$\alpha (\tilde{k} - k) e^{i\tilde{k}\frac{a}{2}} - \beta (k + \tilde{k}) e^{-i\tilde{k}\frac{a}{2}} = 0$$

We get $\beta = \alpha \left(\frac{\tilde{k} - k}{\tilde{k} + k} \right) e^{i\tilde{k}a}$

$$2k f_0 e^{-ik\frac{a}{2}} = \alpha (k + \tilde{k}) e^{-i\tilde{k}\frac{a}{2}} - \alpha \frac{(k - \tilde{k})^2}{(\tilde{k} + k)} e^{i\tilde{k}\frac{3a}{2}}$$

$$\alpha = \frac{2k f_0 e^{-ik\frac{a}{2}}}{(k + \tilde{k}) e^{-i\tilde{k}\frac{a}{2}} - \frac{(k - \tilde{k})^2}{\tilde{k} + k} e^{i\tilde{k}\frac{3a}{2}}}$$

$$\beta = \frac{2k f_0 e^{i\tilde{k}\frac{a}{2}}}{\frac{(k + \tilde{k})^2}{\tilde{k} - k} e^{-i\tilde{k}\frac{3a}{2}} - (\tilde{k} - k) e^{i\tilde{k}\frac{a}{2}}}$$

$$f_T e^{ik\frac{a}{2}} = \alpha \left(e^{i\tilde{k}\frac{a}{2}} + \frac{\tilde{k} - k}{\tilde{k} + k} e^{i\tilde{k}\frac{a}{2}} \right) = \alpha \frac{2\tilde{k}}{\tilde{k} + k} e^{i\tilde{k}\frac{a}{2}}$$

$$f_T = \frac{4k\tilde{k} f_0 e^{-ik}}{(k + \tilde{k})^2 e^{-i\tilde{k}a} - (k - \tilde{k})^2 e^{i\tilde{k}a}}$$

$$f_R = \alpha \left(e^{-i\tilde{k}\frac{a}{2}} + \frac{\tilde{k} - k}{\tilde{k} + k} e^{i\tilde{k}\frac{3a}{2}} \right) e^{-ik\frac{a}{2}} - f_0 e^{-ika}$$

$$= \frac{2k f_0 e^{-ik\frac{a}{2}}}{(k + \tilde{k}) e^{-i\tilde{k}\frac{a}{2}} - \frac{(k - \tilde{k})^2}{\tilde{k} + k} e^{i\tilde{k}\frac{3a}{2}}} \left(e^{-i\tilde{k}\frac{a}{2}} + \frac{\tilde{k} - k}{\tilde{k} + k} e^{i\tilde{k}\frac{3a}{2}} \right) - f_0 e^{-ika}$$

$$= \frac{f_0 e^{-ika}}{(k + \tilde{k}) e^{-i\tilde{k}\frac{a}{2}} - \frac{(k - \tilde{k})^2}{\tilde{k} + k} e^{i\tilde{k}\frac{3a}{2}}} \left[(2k - k - \tilde{k}) e^{-i\tilde{k}\frac{a}{2}} + \frac{k\tilde{k} - k^2 + k^2 - 2k\tilde{k} + \tilde{k}^2}{\tilde{k} + k} e^{i\tilde{k}\frac{3a}{2}} \right]$$

$$f_R = \frac{f_0 e^{-ika}}{(k+\tilde{k})e^{-i\tilde{k}\frac{a}{2}} - \frac{(k-\tilde{k})^2}{k+\tilde{k}} e^{i\tilde{k}\frac{3a}{2}}} \left[(k-\tilde{k}) e^{-i\tilde{k}\frac{a}{2}} + \frac{\tilde{k}(\tilde{k}-k)}{\tilde{k}+k} e^{i\tilde{k}\frac{3a}{2}} \right]$$

Now we have found all the coefficients we require but they include exponential terms which are generically complex.

Initially the sinusoidal wave we have sent was:

$$f_0 \sin(kz - \omega t) = \text{Im} \{ f_0 e^{i(kz - \omega t)} \} \quad \text{where } f_0 \text{ is purely real}$$

But the coefficients we have found are not strictly real. Then in order to obtain the imaginary parts of the reflected and transmitted wave functions, we should compose them as:

$$f_{T-1}(z, t) = \text{Re}\{\alpha\} \sin(\tilde{k}z - \omega t) + \text{Im}\{\alpha\} \cos(\tilde{k}z - \omega t)$$

$$f_{R-1}(z, t) = \text{Re}\{f_R\} \sin(-kz - \omega t) + \text{Im}\{f_R\} \cos(-kz - \omega t)$$

$$f_{R-2}(z, t) = \text{Re}\{\beta\} \sin(-\tilde{k}z - \omega t) + \text{Im}\{\beta\} \cos(-\tilde{k}z - \omega t)$$

$$f_{T-2}(z, t) = \text{Re}\{f_T\} \sin(kz - \omega t) + \text{Im}\{f_T\} \cos(kz - \omega t)$$

keeping in mind that $v = \frac{\omega}{k} = \sqrt{\frac{T}{\mu}}$:

$$k = \omega \sqrt{\frac{\mu_0}{T}} \quad \text{and} \quad \tilde{k} = \omega \sqrt{\frac{\tilde{\mu}}{T}}$$

One can find the steady state solutions everywhere as:

$$f(x, t) = \begin{cases} f_0 \sin(kz - \omega t) + f_{R-1}(z, t) & , \quad z < -a/2 \\ f_{T-1}(z, t) + f_{R-2}(z, t) & , \quad -a/2 < z < a/2 \\ f_{T-2}(z, t) & , \quad z > a/2 \end{cases}$$

What happens if we let $\tilde{\mu} \rightarrow \infty, a \rightarrow 0$ s.t. $\tilde{\mu}a \rightarrow m_0$

Notice that in this limit $ka \propto \sqrt{\mu_0} a \rightarrow 0$

and also $\tilde{k}a \propto \sqrt{\tilde{\mu}} a \rightarrow \sqrt{m_0} \sqrt{a} \rightarrow 0$

Then all the exponential terms in the expressions of the coefficients f_R, α, β, f_T will be equal to 1.

In this limit:

$$\alpha = \frac{2k(k+\tilde{k})f_0}{4k\tilde{k}} = \frac{1}{2} \left(1 + \frac{k}{\tilde{k}}\right) f_0 = \frac{f_0}{2} \quad \text{since } \tilde{k} \rightarrow \infty$$

$$\beta = \frac{2k(\tilde{k}-k)f_0}{4k\tilde{k}} = \frac{1}{2} \left(1 - \frac{k}{\tilde{k}}\right) f_0 = \frac{f_0}{2}$$

$$f_T = \frac{4k\tilde{k}f_0}{4k\tilde{k}} = f_0$$

$$f_R = \frac{(k+\tilde{k})f_0}{4k\tilde{k}} \cdot \frac{k^2 - \tilde{k}^2 + \tilde{k}^2 - k\tilde{k}}{k+\tilde{k}} = \frac{k-\tilde{k}}{\tilde{k}} f_0 = -f_0$$

Then in this limit we have:

$$f(z,t) = \begin{cases} 0 & , z < -\frac{a}{2} \\ f_0 \sin(k\tilde{z}-\omega t) & , -\frac{a}{2} < z < \frac{a}{2} \\ f_0 \sin(kz-\omega t) & , z > \frac{a}{2} \end{cases}$$

*This approach is suggesting that string is stationary for $z < 0$ and oscillating with $f_0 \sin(kz-\omega t)$ for $z > 0$. Obviously, this is nonsense.

From the continuity equations we know that $\frac{\partial \mathcal{E}}{\partial t} + \frac{\partial J}{\partial x} = 0$

→ Clearly $\frac{\partial \mathcal{E}}{\partial t} = 0$ since we are interested in steady state, but $\frac{\partial J}{\partial x} \neq 0$ since we have a net outward flux. Continuity equation doesn't hold

Then this point particle model is not a valid representation for transverse waves. Hence, we can't consider this limit,